Near-Perfect but Misleading: Re-evaluating Machine Learning for Recurrence Prediction in Differentiated Thyroid Cancer

Anonymous authors
Paper under double-blind review

Abstract

Machine-learning studies on a widely used public dataset of 383 patients with differentiated thyroid cancer (UCI repository) report accuracies close to 99% for predicting recurrence. We examine where these results come from. A data audit shows that one variable, the response to initial therapy, which is assessed during follow-up, classifies 94.5% of patients correctly on its own (AUC 0.979), and confirms 19 duplicate records. Starting from an earlier analysis that reported 98.7% accuracy, we compare four evaluation protocols, from a single split with hyperparameters tuned on the test set to repeated nested cross-validation. Tuning on the test set reproduces the 98.7%; tuning inside the training set gives 0.974 on the same split, and accuracy over 50 random splits ranges from 0.896 to 1.000. Under nested cross-validation, five classifiers reach AUCs of 0.961–0.987, but none is significantly better than the response to therapy used alone (0.978), which needs no training. Without that variable, the AUC falls to 0.931 (logistic regression) and 0.934 (random forest), and sensitivity falls from about 0.90 to 0.680–0.763. Calibration, decision-curve and SHAP analyses support the same conclusion. Near-perfect results on this benchmark mainly reflect a post-treatment predictor and optimistic evaluation; we recommend stating the time of prediction, reporting single-variable baselines and using nested evaluation.

1 Introduction

Thyroid cancer is the most common endocrine malignancy, and its incidence has risen in many countries over recent decades (Wiltshire et al., 2016). Most cases are differentiated thyroid cancers (DTC), mainly papillary and follicular carcinomas, and their prognosis is good. Recurrence is nevertheless frequent: depending on the definition and the length of follow-up, it affects roughly 5% to 30% of patients (Grogan et al., 2013; van de Berg et al., 2024). Predicting which patients will recur would help adapt the intensity of follow-up.

A public dataset released by Borzooei et al. (2024) on the UCI Machine Learning Repository (Borzooei & Tarokhian, 2023), with 383 patients and 16 clinicopathological variables, has made this problem easy to study. Many studies have since trained classical machine-learning models and neural networks on it, and most of them report accuracies or areas under the ROC curve (AUC) above 0.96 (Öztürk et al., 2025; Kumari et al., 2026; Thakur et al., 2026; Clark et al., 2024; Al-Sayed Ahmad & Haddad, 2024). Their interpretability analyses agree that the most important variable is the *response to initial therapy*, followed by the ATA risk category.

These results deserve a closer look for three reasons. First, results on small datasets are sensitive to the evaluation protocol: a single train/test split gives a noisy estimate, and choosing hyperparameters on the test set biases it upwards (Varma & Simon, 2006; Cawley & Talbot, 2010). Second, data leakage, where information about the test set or the outcome reaches the model during training, is a common cause of over-optimistic results in machine-learning-based science (Kaufman et al., 2012; Kapoor & Narayanan, 2023). Third, the response to therapy is assessed during follow-up, months after the initial treatment, and its worst category (*structural incomplete*) means that disease is still visible on imaging (Tuttle et al., 2010; Haugen et al., 2016). A model that relies on it is partly using a description of the outcome as a predictor.

Our starting point is an earlier, unpublished analysis of our own, which we call the *original analysis* throughout: a support vector machine (SVM) with Chi-square and random-forest feature selection, evaluated on a single stratified 80/20 split, reached an accuracy of 98.7% and an AUC of 0.997. When we re-examined it, we found that it shared all three weaknesses. In this paper we re-evaluate the problem with a stricter protocol, and use it as a case study of how a small public clinical table can produce near-perfect but misleading benchmark results. The lessons are not specific to thyroid cancer: post-treatment variables, duplicated records and single-split evaluation are common in public tabular health datasets. Our contributions are:

1. a data audit showing that one variable, *Response*, almost determines the outcome on its own, and confirming the duplicate records noted in earlier work that has not been peer reviewed (Aggarwal et al., 2024; Ashraf & Ilyas, 2026);
2. a step-by-step decomposition of how test-set tuning and the choice of a single split inflate the reported performance;
3. a repeated nested cross-validation of five classifiers, compared with a reference that uses *Response* alone and needs no training, which to our knowledge no previous study has reported;
4. an ablation without *Response*, which estimates what can be predicted from the information available at the end of initial treatment, together with calibration, decision-curve and SHAP analyses.

The rest of the paper is organised as follows. Section 2 reviews prior work on this dataset. Section 3 describes the data, the audit and the evaluation protocols. Section 4 presents the results, Section 5 discusses their implications and limitations, and Section 6 concludes.

2 Related work

2.1 Studies on the UCI dataset

Table 1 summarises the studies of recurrence on the UCI data. Borzooei et al. (2024), who collected the cohort, trained several classifiers on a single split of 283 training and 100 validation patients; the SVM performed best, with an AUC of 0.997, and they also built a variant without the ATA risk category. Öztürk et al. (2025) used one round of five-fold cross-validation after recursive feature elimination; the RF reached an accuracy of 0.974 and an AUC of 0.993, and SHAP ranked response to treatment and ATA risk first. Thakur et al. (2026) reported an RF accuracy of 0.983 and an AUC of 1.00 on a single 70/30 split, together with a nested cross-validation (accuracy about 0.96); their Gini and SHAP rankings placed *Response* first, and they noted that an incomplete response reflects residual disease. Clark et al. (2024) reported an RF accuracy of 0.99 and an AUC of 0.996 on a single 80/20 split, with *Response* dominating feature importance. Kumari et al. (2026) compared eleven classifiers and Bayesian neural networks on an 80/20 split; the best network reached an accuracy of 0.974, and SHAP again ranked *Response*, *Risk* and *Stage* first. Al-Sayed Ahmad & Haddad (2024) trained a neural network on an 80/20 split and reported a test accuracy of 0.96, with *Response* and *Stage* as the main variables in their Morris sensitivity analysis. Hu et al. (2026) took a different approach: they prioritised variables available at diagnosis, and *Response* does not appear among their reported predictors. Their RF reached an AUC of 0.931, clearly below the other studies.

Two points are shared by almost all of these studies: *Response* is the dominant predictor whenever it is used, and the headline result comes from a single split or a single round of cross-validation. None of the peer-reviewed studies in Table 1 reports the duplicate records or compares the models with *Response* used on its own. One study states explicitly that the dataset contains no duplicates (Shrestha et al., 2025).

Two pieces of work that have not been peer reviewed anticipate parts of our audit. A student research report removed the duplicates and analysed 364 patients (Aggarwal et al., 2024), and a recently submitted study of robustness to label poisoning also removed the 19 duplicates and excluded *Response* as a post-treatment variable (Ashraf & Ilyas, 2026). Outside the recurrence task, Khwanmuang et al. (2025), who predicted the ATA risk class, built one scenario without *Response* to restrict the model to preoperative predictors. Our study builds on these observations: it quantifies how much *Response* alone explains, how the evaluation protocol inflates the estimates, and what remains once *Response* is removed.

Table 1: Machine-learning studies of recurrence on the UCI differentiated thyroid cancer data, as reported by their authors. n.r.: not reported in the parts of the paper we could access.

Study	Evaluation	Uses *Response*	Best reported result	Top predictors
Borzooei et al. (2024)	Single split (283/100)	n.r.	SVM, AUC 0.997	n.r.
Öztürk et al. (2025)	One round of 5-fold CV	Yes	RF, accuracy 0.974, AUC 0.993	Response, Risk, T, Age
Thakur et al. (2026)	Single 70/30 split; also nested CV	Yes	RF, accuracy 0.983, AUC 1.00 (split)	Response, Risk, N, T
Hu et al. (2026)	Single 70/30 split, CV within training set	Not among reported predictors	RF, AUC 0.931	Risk, N, T, Age
Clark et al. (2024)	Single 80/20 split	Yes	RF, accuracy 0.99, AUC 0.996	Response, Risk
Kumari et al. (2026)	Single 80/20 split	Yes	Bayesian NN, accuracy 0.974	Response, Risk, Stage
Al-Sayed Ahmad & Haddad (2024)	Single 80/20 split	Yes	Neural network, test accuracy 0.96	Response, Stage
Aggarwal et al. (2024)[a]	Single 75/25 split, CV tuning; duplicates removed ($n = 364$)	Yes	RF, accuracy 0.945	–
Ashraf & Ilyas (2026)[a]	3 seeds × 5-fold CV; duplicates removed ($n = 364$)	Excluded (post-treatment)	Robustness to label poisoning	–

[a] Not peer reviewed (student research report; submitted manuscript with public code).

2.2 Studies on other cohorts

Other studies used larger or different cohorts. Alam et al. (2024) found that logistic regression slightly outperformed RF. Wang et al. (2024) selected predictors with LASSO and used five-fold cross-validation; RF performed best, with thyroglobulin and the lymph-node ratio as the top variables. Park & Lee (2021) found that a decision tree reached an accuracy of 0.95, and Li et al. (2025) reported an RF accuracy of 0.808 on 2,388 intermediate- and high-risk patients. Das et al. (2025) built a stacking classifier on RF and gradient boosting, and Atay et al. (2024) combined association rules with classification on 550 patients. The lower accuracies reported on these larger cohorts, which do not all include a response-to-therapy variable, are a first hint that the near-perfect scores on the UCI dataset may not transfer.

2.3 Evaluation pitfalls in clinical machine learning

Selecting hyperparameters or features on the same data used to estimate performance produces optimistic estimates, and the bias is largest on small datasets (Varma & Simon, 2006; Cawley & Talbot, 2010). Nested cross-validation avoids it by tuning inside each training fold. More generally, Kaufman et al. (2012) and Kapoor & Narayanan (2023) describe how leakage, including features that would not be available at prediction time, has inflated results across many fields. Reporting guidelines such as TRIPOD+AI (Collins et al., 2024) ask authors to state when each predictor is measured relative to the moment of prediction, and to report calibration and clinical utility in addition to discrimination (Van Calster et al., 2019; Vickers & Elkin, 2006).

3 Materials and methods

The study follows the TRIPOD+AI reporting guideline for prediction-model studies (Collins et al., 2024) where it applies to a secondary analysis of public data.

3.1 Data

We used the *Differentiated Thyroid Cancer Recurrence* dataset from the UCI Machine Learning Repository (Borzooei & Tarokhian, 2023), collected by Borzooei et al. (2024). It contains 383 patients treated for

differentiated thyroid cancer at a single centre and followed for 10 to 15 years. Each record has 16 clinicopathological variables and a binary outcome, *Recurred* (108 yes, 275 no). The variables are listed in Table 2. The data are de-identified and publicly available, so no ethical approval was required.

Table 2: Variables of the UCI differentiated thyroid cancer dataset (all 383 records). Counts are given for each level.

Variable	Type	Levels (count)
Age	Integer	15–82 years (median 37)
Gender	Binary	F (312), M (71)
Smoking	Binary	No (334), Yes (49)
Hx Smoking	Binary	No (355), Yes (28)
Hx Radiotherapy	Binary	No (376), Yes (7)
Thyroid function	Nominal	Euthyroid (332), clinical hyperthyroidism (20), subclinical hypothyroidism (14), clinical hypothyroidism (12), subclinical hyperthyroidism (5)
Physical examination	Nominal	Multinodular goiter (140), single nodular goiter right (140) / left (89), normal (7), diffuse goiter (7)
Adenopathy	Nominal	No (277), right (48), bilateral (32), left (17), extensive (7), posterior (2)
Pathology	Nominal	Papillary (287), micropapillary (48), follicular (28), Hürthle cell (20)
Focality	Binary	Uni-focal (247), multi-focal (136)
Risk (ATA initial)	Ordinal	Low (249), intermediate (102), high (32)
T	Ordinal	T1a (49), T1b (43), T2 (151), T3a (96), T3b (16), T4a (20), T4b (8)
N	Ordinal	N0 (268), N1a (22), N1b (93)
M	Ordinal	M0 (365), M1 (18)
Stage	Ordinal	I (333), II (32), III (4), IVA (3), IVB (11)
Response to therapy	Ordinal	Excellent (208), indeterminate (61), biochemical incomplete (23), structural incomplete (91)
Recurred (outcome)	Binary	No (275), Yes (108)

Two variables deserve attention. *Risk* is the initial American Thyroid Association (ATA) risk of recurrence, assigned from surgical and pathological findings (Haugen et al., 2016). *Response* is the response to initial therapy (excellent, indeterminate, biochemical incomplete, structural incomplete). In the ATA framework it is assessed during follow-up, within the first two years after surgery and radioiodine, and is used to re-stage the patient (Tuttle et al., 2010; Haugen et al., 2016). A *structural incomplete* response means that persistent or recurrent disease is visible on imaging. *Response* is therefore not available at the time of initial treatment, and part of it describes the outcome being predicted.

3.2 Data audit

Before modelling, we checked the data for exact duplicate records (identical values on all 17 columns) and tabulated the outcome against each level of *Response*. We also counted how many test patients in the original 80/20 split (stratified, `random_state`=42) had an identical record in the training set.

3.3 Encoding and modelling pipeline

Ordinal variables (*Risk*, *T*, *N*, *M*, *Stage*, *Response*) were coded as integers in their clinical order. Binary variables were coded 0/1. The four nominal variables (*Thyroid function*, *Physical examination*, *Adenopathy*, *Pathology*) were one-hot encoded, giving 32 input columns. These steps use no statistics estimated from the data.

Every model was wrapped in a single scikit-learn pipeline (Pedregosa et al., 2011): one-hot encoding, Chi-square feature selection (`SelectKBest`, $k \in \{10, 20, \text{all}\}$), standardisation (for LR, SVM and MLP), and the classifier. Because the pipeline is refitted on every training set, feature selection and scaling never see the corresponding test data.

We compared five classifiers: L2-regularised logistic regression (LR), decision tree (DT), random forest (RF, 200 trees), support vector machine with an RBF kernel (SVM), and a multilayer perceptron (MLP). Their hyperparameter grids are given in Table 3. For RF, k was fixed to all features to limit computation.

Table 3: Hyperparameter grids searched by 5-fold inner cross-validation (AUC). k is the number of columns kept by Chi-square selection.

Model	Grid
LR	$C \in \{0.01, 0.1, 1, 10\}$; $k \in \{10, 20, \text{all}\}$
DT	max depth $\in \{2, 3, 4, 5, \text{none}\}$; min. samples per leaf $\in \{1, 5, 10\}$; $k \in \{10, 20, \text{all}\}$
RF	200 trees; max depth $\in \{\text{none}, 5\}$; min. samples per leaf $\in \{1, 3\}$; $k = \text{all}$
SVM	RBF kernel; $C \in \{0.1, 1, 10, 50, 100\}$; $\gamma \in \{\text{scale}, 0.01, 0.1\}$; $k \in \{10, 20, \text{all}\}$
MLP	hidden layers $\in \{(32), (64, 32)\}$; L2 penalty $\alpha \in \{10^{-2}, 1\}$; $k \in \{10, 20, \text{all}\}$

As a reference that requires no training, we scored each patient by the ordinal *Response* code alone. The AUC of this score measures how well *Response* ranks patients by itself. For threshold metrics, we used the rule "predict recurrence if and only if the response is structural incomplete".

3.4 Evaluation protocols

To separate the sources of optimism in the original estimate, we evaluated the SVM and RF under four protocols of increasing strictness:

(A) **Single split, tuned on the test set.** The original seed-42 split. Every configuration in the grid is evaluated on the test set and the best test AUC is reported. This reproduces the original hyperparameter-sensitivity procedure.

(B) **Single split, tuned in the training set.** Same split. Hyperparameters are chosen by 5-fold cross-validation inside the training set, and the test set is used once.

(C) **Fifty random splits.** Protocol B repeated over 50 stratified 80/20 splits with different seeds, to show how much a single split can vary.

(D) **Repeated nested cross-validation.** Five repeats of stratified 5-fold cross-validation (25 outer test folds). Inside each outer training fold, a 5-fold grid search on AUC selects the hyperparameters and k; the selected pipeline is refitted on the outer training fold and evaluated once on the outer test fold (Varma & Simon, 2006; Cawley & Talbot, 2010).

Protocol D was the primary analysis and was applied to all five models. It was run on the 364 unique records (primary) and on all 383 records (sensitivity analysis).

3.5 Ablation

To measure how much of the performance depends on *Response*, we repeated Protocol D with three feature sets: all 16 variables (*Full*); all variables except *Response* (*No Response*), which corresponds to the information available at the end of initial treatment; and all variables except *Response* and *Risk* (*No Response, no Risk*), since *Risk* is itself a summary of T, N, M and pathological findings.

3.6 Performance measures and statistical comparison

On each outer test fold we computed the AUC, accuracy, balanced accuracy, sensitivity, specificity, F1-score and, for models that output probabilities, the Brier score. We report the mean and standard deviation over the 25 outer folds. Threshold metrics use the default decision threshold of each classifier (0.5 for probabilities, 0 for the SVM decision function).

Fold-level results from repeated cross-validation are not independent, because training sets overlap. We therefore compared AUCs with the corrected resampled t-test of Nadeau & Bengio (2003), which inflates the variance by a factor $(1/J + n_{\text{test}}/n_{\text{train}})$, with $J = 25$ folds and $n_{\text{test}}/n_{\text{train}} = 1/4$. We tested each model against the *Response*-only score, the best model against the others, and the *Full* against the *No Response* feature set for each model.

3.7 Calibration and clinical utility

For the probabilistic models we pooled the out-of-fold predictions of the first cross-validation repeat (each patient predicted once) and computed the Brier score, the calibration slope and the calibration-in-the-large (Van Calster et al., 2019). We plotted calibration curves (eight quantile bins) and decision curves (Vickers & Elkin, 2006) for thresholds between 5% and 60%, comparing each model with the "treat all" and "treat none" strategies.

3.8 Interpretation

To see which variables the models rely on, we fitted the RF on all 364 unique patients (hyperparameters by 5-fold cross-validation) with and without *Response*, and computed SHAP values with the tree explainer (Lundberg & Lee, 2017). For one-hot encoded variables, the mean absolute SHAP values of the indicator columns were summed back to the original variable.

3.9 Software and reproducibility

Analyses used Python 3.13, scikit-learn 1.9, SHAP and statsmodels. All random seeds are fixed. The notebook that produces every number, table and figure in this paper is provided as anonymised supplementary material and will be released publicly upon acceptance.

4 Results

4.1 Data audit

As previously noted by Aggarwal et al. (2024) and Ashraf & Ilyas (2026), the dataset contains 19 exact duplicate records: 35 rows share their complete profile with at least one other row. All of them are women without recurrence, with an *excellent* response to therapy and a low-risk profile. Removing the duplicates leaves 364 unique records (108 recurrences, 256 non-recurrences). With only 16 coarse variables, some of these rows may be different patients who happen to share the same values, so we treat deduplication as a sensitivity choice rather than a correction. In the original seed-42 split, 7 of the 77 test patients had an identical record in the training set.

Table 4 shows the outcome by response to therapy. Recurrence occurred in 1 of 208 patients with an excellent response (0.5%) and in 89 of 91 patients with a structural incomplete response (97.8%). The rule "recurrence if and only if the response is structural incomplete" classifies 362 of the 383 records correctly (accuracy 94.5%, sensitivity 82.4%, specificity 99.3%), and the ordinal response code alone gives an AUC of 0.979. No model is involved.

Table 4: Recurrence (Rec.) by response to initial therapy (all 383 records).

Response to therapy	No rec.	Rec.	Rate
Excellent	207	1	0.5%
Indeterminate	54	7	11.5%
Biochemical Incomplete	12	11	47.8%
Structural Incomplete	2	89	97.8%

4.2 How the evaluation protocol changes the estimate

Table 5 decomposes the original estimate. With the seed-42 split and hyperparameters chosen on the test set (Protocol A), our pipeline reproduces the original accuracy: the SVM reaches 0.987 (AUC 1.000, one false negative). When the hyperparameters are chosen inside the training set (Protocol B), the same split gives an accuracy of 0.974 and an AUC of 0.995. On this split, *Response* alone already reaches an accuracy of 0.935 and an AUC of 0.986.

Table 5: Test-set performance of the SVM and RF under increasingly strict protocols (all 383 records, full feature set). Rows A and B use the original seed-42 split (77 test patients). Row C is the mean over 50 random 80/20 splits. The first row gives the figures reported in the original analysis.

Model	Protocol	Accuracy	AUC	Sensitivity	Specificity
SVM	Original analysis (seed-42 split, tuned on the test set)	0.987	0.997	0.955	1.000
SVM	(A) Seed-42 split, tuned on the test set	0.987	1.000	0.955	1.000
SVM	(B) Seed-42 split, tuned in training folds	0.974	0.995	0.909	1.000
SVM	(C) 50 random splits, tuned in training folds (mean)	0.956	0.984	0.904	0.976
RF	(A) Seed-42 split, tuned on the test set	0.961	0.991	0.864	1.000
RF	(B) Seed-42 split, tuned in training folds	0.961	0.990	0.864	1.000
RF	(C) 50 random splits, tuned in training folds (mean)	0.959	0.982	0.891	0.986
Response alone	Seed-42 split, no training	0.935	0.986	0.773	1.000

Figure 1 shows the spread over 50 random splits (Protocol C). The SVM accuracy ranged from 0.896 to 1.000 (mean 0.956 $\pm$ 0.021), and the RF accuracy from 0.922 to 1.000 (mean 0.959 $\pm$ 0.018). Only 5 of the 50 SVM splits reached the originally reported accuracy of 0.987. With 77 test patients, one misclassification changes the accuracy by 1.3 percentage points, so the differences between models reported on a single split are within the noise.

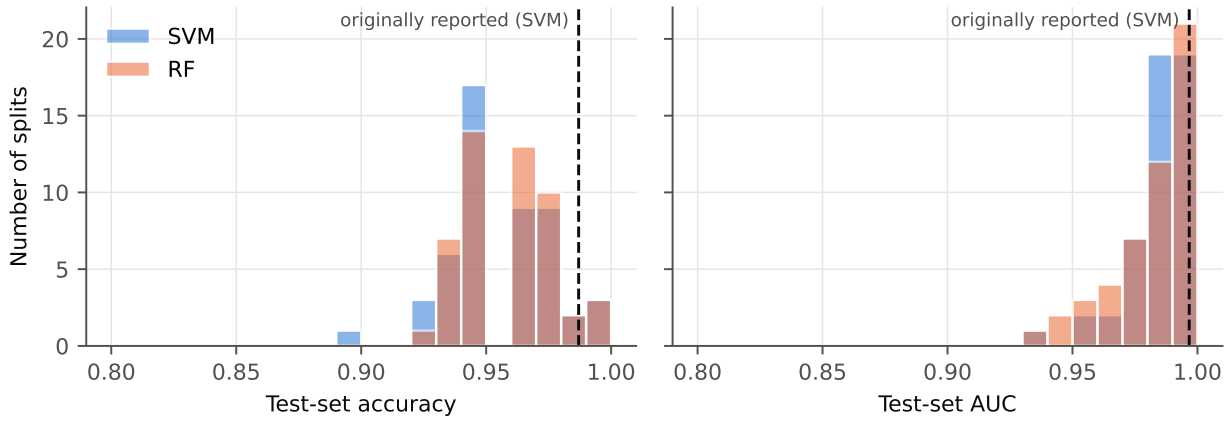


Figure 1: Test-set accuracy (left) and AUC (right) of the SVM and RF over 50 random stratified 80/20 splits, with hyperparameters tuned inside each training set. The dashed line marks the value reported in the original analysis.

4.3 Repeated nested cross-validation

Table 6 gives the nested cross-validation results on the 364 unique records with all variables. The mean AUCs range from 0.961 (decision tree) to 0.987 (logistic regression), and the *Response* code alone reaches 0.978 $\pm$ 0.015. The gain of the trained models over this reference is at most 0.009 AUC points, and none

of the five models differs significantly from it (Table 7; all $p > 0.05$); the decision tree is lower on average. The four best models cannot be separated either: the logistic regression differs significantly only from the decision tree. Where the trained models do help is sensitivity: apart from the decision tree, they detect about 90% of recurrences, against 0.824 for the *structural incomplete* rule, at the cost of a slightly lower specificity.

Table 6: Repeated nested cross-validation on the 364 unique records, full feature set (mean ± SD over 25 outer test folds).

Model	AUC	Balanced acc.	Sensitivity	Specificity	Brier
Response alone[a]	0.978 ± 0.015	0.908 ± 0.033	0.824 ± 0.064	0.992 ± 0.010	–
Logistic regression	0.987 ± 0.010	0.943 ± 0.027	0.900 ± 0.052	0.985 ± 0.018	0.035 ± 0.012
Decision tree	0.961 ± 0.027	0.913 ± 0.038	0.854 ± 0.071	0.973 ± 0.034	0.048 ± 0.020
Random forest	0.984 ± 0.016	0.939 ± 0.032	0.893 ± 0.063	0.986 ± 0.017	0.042 ± 0.011
SVM (RBF)	0.986 ± 0.011	0.941 ± 0.031	0.902 ± 0.061	0.980 ± 0.021	–
Multilayer perceptron	0.983 ± 0.012	0.939 ± 0.028	0.905 ± 0.053	0.973 ± 0.016	0.037 ± 0.014

[a] Ordinal response code used as the score (AUC); threshold metrics use the rule "recurrence if and only if structural incomplete". No training. The SVM has no Brier score because it outputs decision values, not probabilities.

Table 7: Pairwise AUC comparisons on the full feature set (corrected resampled t-test, 25 outer folds).

Comparison	Mean ΔAUC	t	p
LR vs Response alone	+0.009	1.90	0.070
DT vs Response alone	-0.017	-1.63	0.116
RF vs Response alone	+0.006	0.74	0.468
SVM vs Response alone	+0.009	1.31	0.204
MLP vs Response alone	+0.005	0.90	0.378
LR vs DT	+0.026	2.20	0.038
LR vs RF	+0.003	0.49	0.626
LR vs SVM	+0.000	0.11	0.917
LR vs MLP	+0.004	0.84	0.408

The sensitivity analysis on all 383 records, duplicates included, gives very similar figures (RF AUC 0.984, SVM 0.987, LR 0.985). Because the duplicates are all easy negatives, keeping them changes the estimates little.

4.4 Ablation of the post-treatment variable

Removing *Response* lowers the AUC of every model (Table 8 and Figure 2). Without it, the best models reach an AUC of about 0.931 (LR) and 0.934 (RF), and the drop is significant for all of them. Removing *Risk* as well lowers the AUC further, to 0.915 for LR and 0.924 for RF. Sensitivity is affected most: on the full feature set the RF detects 0.893 of recurrences on average, against 0.763 without *Response*.

Table 8: AUC (mean ± SD over 25 outer folds) with and without the post-treatment variable *Response*, 364 unique records. Δ and p compare the full and the No-Response feature sets (corrected resampled t-test).

Model	Full	No Response	No Response, no Risk	ΔAUC	p
Logistic regression	0.987 ± 0.010	0.931 ± 0.028	0.915 ± 0.036	+0.055	<0.001
Decision tree	0.961 ± 0.027	0.902 ± 0.038	0.862 ± 0.054	+0.059	0.021
Random forest	0.984 ± 0.016	0.934 ± 0.027	0.924 ± 0.027	+0.050	0.001
SVM (RBF)	0.986 ± 0.011	0.930 ± 0.027	0.913 ± 0.032	+0.056	<0.001
Multilayer perceptron	0.983 ± 0.012	0.923 ± 0.030	0.900 ± 0.049	+0.060	<0.001

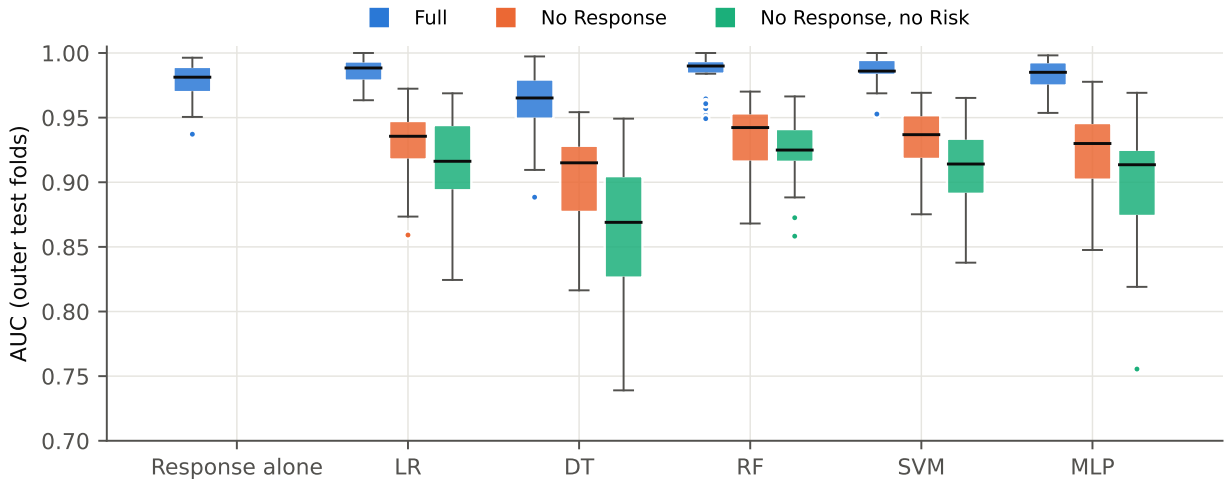


Figure 2: AUC on the 25 outer test folds of repeated nested cross-validation (364 unique records) for each model and feature set. *Response* alone requires no training and is only defined for the full feature set.

4.5 Calibration and clinical utility

Table 9 and Figure 3 report calibration. With all variables, the logistic regression has the lowest Brier score (0.035) and a calibration slope of 1.21. The random forest has a slope of 1.86, meaning that its probabilities are too close to the middle, and the decision tree a slope of 0.33, meaning that they are too extreme. Without *Response*, the Brier scores are two to three times higher (LR 0.098, RF 0.091). With only 108 events, these slopes are imprecise and should be read as indications only (Riley et al., 2020). In the decision-curve analysis, the logistic regression gives a higher net benefit than treating all or no patients across the 5%–60% threshold range, with or without *Response*; the gap between the two curves is the value that *Response* adds once it is known.

Table 9: Calibration of the probabilistic models (pooled out-of-fold predictions, first repeat, 364 unique records). CITL: calibration-in-the-large. Ideal values: slope 1, CITL 0.

Features	Model	Brier	Slope	CITL
Full	LR	0.035	1.21	+0.08
Full	DT	0.046	0.33	+0.24
Full	RF	0.043	1.86	+0.02
Full	MLP	0.046	0.74	+0.24
No Response	LR	0.098	1.68	-0.03
No Response	DT	0.102	0.42	+0.07
No Response	RF	0.091	1.28	+0.06
No Response	MLP	0.096	0.81	+0.09
No Response, no Risk	LR	0.107	1.57	+0.03
No Response, no Risk	DT	0.128	0.32	+0.26
No Response, no Risk	RF	0.099	1.14	+0.07
No Response, no Risk	MLP	0.114	0.69	+0.09

4.6 Which variables the models use

With all variables, the SHAP analysis of the RF (Figure 4) ranks Response first (mean |SHAP| 0.167), 2.6 times the second variable, Risk (0.065). This agrees with every previous study on this dataset. Without *Response*, the importance is spread over Risk (0.105), Adenopathy (0.086), N (0.054) and T (0.039), which, unlike *Response*, are normally available by the end of initial treatment.

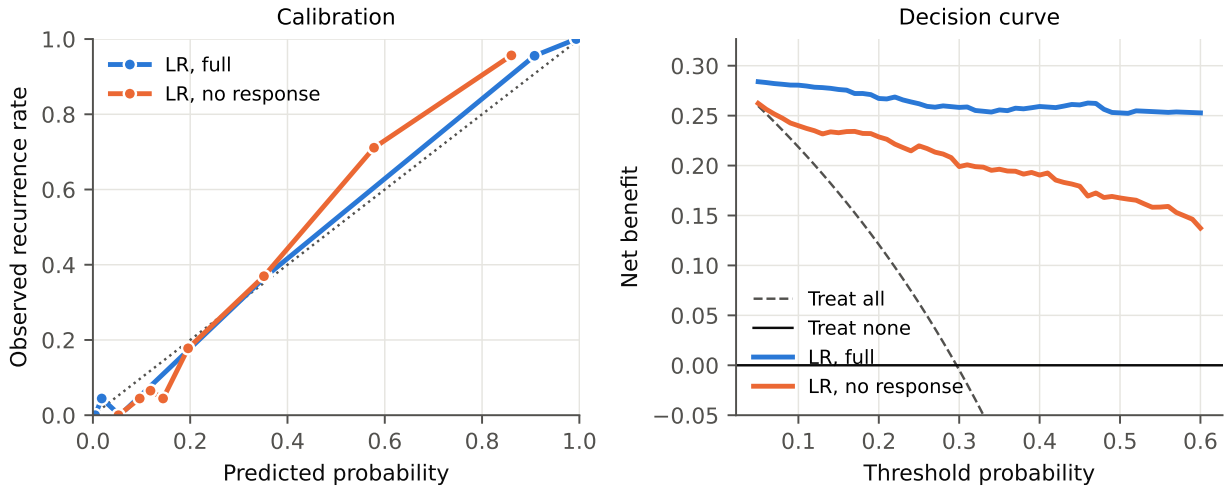


Figure 3: Calibration curves (left, eight quantile bins) and decision curves (right) of the logistic regression with and without *Response*, from pooled out-of-fold predictions on the 364 unique records.

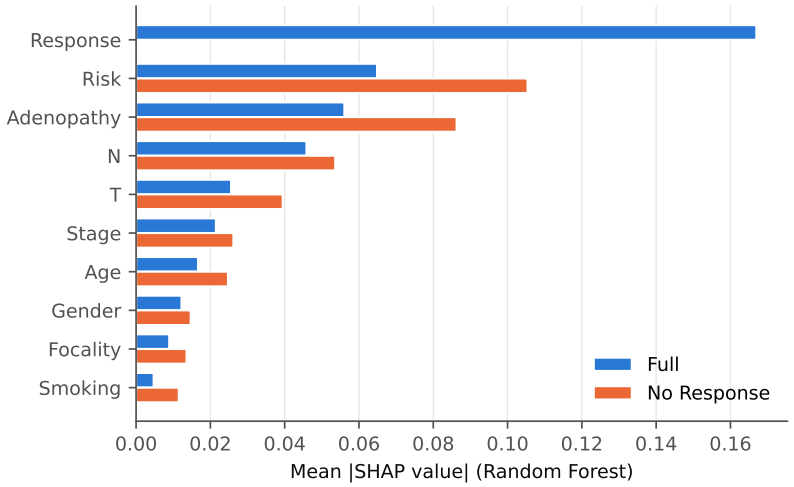


Figure 4: Mean absolute SHAP values of the random forest fitted on the 364 unique records, with and without *Response*. One-hot columns are summed back to their original variable.

5 Discussion

5.1 Principal findings

On the UCI differentiated thyroid cancer dataset, near-perfect results are easy to obtain and hard to interpret. Three factors explain most of them. First, choosing hyperparameters on the test set and reporting a single favourable split raise the apparent accuracy; under nested cross-validation the same models are still very good, but no longer separable from one another. Second, one variable does most of the work: the response to therapy alone reaches an AUC of 0.978 under cross-validation, and the trained models add at most 0.009 AUC points to it, a difference that is not statistically significant. Third, that variable is measured during follow-up, and its worst category largely describes persistent disease. Without it, the AUC falls to about 0.931, which is a more realistic estimate of what can be predicted from information available at the end of initial treatment.

The duplicate records, already noted by Aggarwal et al. (2024) and Ashraf & Ilyas (2026), matter less than we expected. 7 of them sat on both sides of the original split, but they are all easy negatives, so removing them changes the cross-validated estimates very little.

5.2 Why *Response* dominates

In the ATA guidelines, the response to therapy is the basis of *dynamic risk stratification*: the initial risk estimate is revised during the first years of follow-up, using thyroglobulin levels and imaging (Tuttle et al., 2010; Haugen et al., 2016). A structural incomplete response means that disease is visible on imaging, which is close to the clinical definition of persistent or recurrent disease (van de Berg et al., 2024). In this dataset, 89 of the 91 patients with that response are labelled as recurrences. A model that uses *Response* is therefore not predicting recurrence at diagnosis; it is closer to confirming a finding already made during follow-up.

This does not make *Response* useless. It is a legitimate predictor if the model is meant to be used after the first follow-up assessment, to refine the risk of later events. But the prediction time must then be stated, and the comparison should be with the clinical response category itself, which clinicians already use. The SHAP rankings reported in earlier studies, which all place *Response* first (Öztürk et al., 2025; Thakur et al., 2026; Kumari et al., 2026; Al-Sayed Ahmad & Haddad, 2024), are best read as a confirmation of the ATA framework rather than as a new finding. Consistently, the one study that restricted itself to variables available at diagnosis and does not report *Response* among its predictors obtained an AUC of 0.931 (Hu et al., 2026), very close to our estimates without *Response* (0.931 for LR, 0.934 for RF).

5.3 Implications for future studies on this dataset

We suggest five practices for future work on this dataset and similar small clinical tables:

1. state the moment of prediction, and report results without post-treatment variables such as *Response* when the model is meant for use at the end of initial treatment;
2. report the performance of *Response* alone, so that readers can see what the model adds;
3. tune hyperparameters and select features inside the training folds (nested cross-validation), never on the test set;
4. report the variability over repeated splits or folds, and test differences between models with a method that accounts for overlapping training sets (Nadeau & Bengio, 2003);
5. check for duplicate records and report calibration and clinical utility, as recommended by TRIPOD+AI (Collins et al., 2024).

5.4 Limitations

This study has the limitations of the data it re-analyses. The cohort comes from a single centre and is small, with 108 recurrences; external validation was not possible. The ablation without *Response* is a more

realistic setting, but it was not designed prospectively, and other variables in the dataset (for example adenopathy found during follow-up) may also have been recorded after the initial treatment; the dataset documentation does not say. We could not tell whether duplicate records are different patients or repeated entries. The decision curves use the outcome prevalence of this cohort, which may differ elsewhere. Finally, our hyperparameter grids are modest, and other models or grids could do slightly better, although the ceiling set by *Response* alone leaves little room for improvement on the full feature set.

6 Conclusion

The accuracies close to 99% reported for recurrence prediction on the UCI differentiated thyroid cancer dataset mostly reflect one post-treatment variable and favourable evaluation choices. Under repeated nested cross-validation, the response to therapy alone performs almost as well as any trained model, and without it the AUC is about 0.931. Future studies on this dataset should state when the prediction is made, compare their models with the response to therapy alone, and use nested evaluation. External, multi-centre cohorts with clearly timed predictors are needed before any of these models can support clinical decisions.

Broader Impact Statement

This work re-analyses public, de-identified data and proposes no model for clinical use. Its main risk runs the other way: published models that appear near-perfect on this dataset could be mistaken for decision-support tools. By showing that their performance rests largely on a post-treatment variable, we hope to reduce that risk. Any clinical use of recurrence models would require prospective, multi-centre validation with predictors timed to the moment of decision.

Data Availability

The dataset is publicly available from the UCI Machine Learning Repository (Borzooei & Tarokhian, 2023) under a CC BY 4.0 licence. Code is provided as supplementary material.

References

Anay Aggarwal, Marly Gotti, Ekam Kaur, and Susie Lu. Comparative analysis of machine learning models for thyroid cancer recurrence prediction. MIT PRIMES research paper, 2024. URL `https://math.mit.edu/research/highschool/primes/materials/2024/Aggarwal-Gotti-Kaur-Lu.pdf`.

Mohammad Al-Sayed Ahmad and Jude Haddad. An explainable AI model for predicting the recurrence of differentiated thyroid cancer. In *2024 Second Jordanian International Biomedical Engineering Conference (JIBEC)*, pp. 84–89, 2024. doi: 10.1109/JIBEC63210.2024.10932125.

Shah Alam, Mohammad Abir Hider, Abdullah Al Mukaddim, Farhana Rahman Anonna, Md Sazzad Hossain, Md Khalilor Rahman, and Md Nasiruddin. Machine learning models for predicting thyroid cancer recurrence: A comparative analysis. *Journal of Medical and Health Studies*, 5(4):113–129, 2024. doi: 10.32996/jmhs.2024.5.4.14.

Mobeen Muhammad Ashraf and Iqra Ilyas. Hidden recurrences: Robustness of thyroid cancer recurrence predictors to targeted training-label poisoning. GitHub repository, 2026. URL `https://github.com/MobeenMuhammadAshraf/dtc-poisoning`. Manuscript submitted; code repository.

Feyza Firat Atay, Fatma Hilal Yagin, Cemil Colak, Emin Tamer Elkiran, Nasrin Mansuri, Fuzail Ahmad, and Luca Paolo Ardigò. A hybrid machine learning model combining association rule mining and classification algorithms to predict differentiated thyroid cancer recurrence. *Frontiers in Medicine*, 11:1461372, 2024. doi: 10.3389/fmed.2024.1461372.

Shiva Borzooei and Aidin Tarokhian. Differentiated thyroid cancer recurrence. UCI Machine Learning Repository, dataset ID 915, 2023. URL `https://archive.ics.uci.edu/dataset/915/differentiated+thyroid+cancer+recurrence`. Donated 30 October 2023; 383 instances; CC BY 4.0.

Shiva Borzooei, Giovanni Briganti, Mitra Golparian, Jerome R. Lechien, and Aidin Tarokhian. Machine learning for risk stratification of thyroid cancer patients: a 15-year cohort study. *European Archives of Oto-Rhino-Laryngology*, 281(4):2095–2104, 2024. doi: 10.1007/s00405-023-08299-w. Published online 30 October 2023.

Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11(70):2079–2107, 2010. URL `https://www.jmlr.org/papers/v11/cawley10a.html`.

Elizabeth Clark, Samantha Price, Theresa Lucena, Bailey Haberlein, Abdullah Wahbeh, and Raed Seetan. Predictive analytics for thyroid cancer recurrence: A machine learning approach. *Knowledge*, 4(4):557–570, 2024. doi: 10.3390/knowledge4040029.

Gary S. Collins, Karel G. M. Moons, Paula Dhiman, Richard D. Riley, Andrew L. Beam, Ben Van Calster, Marzyeh Ghassemi, Xiaoxuan Liu, Johannes B. Reitsma, Maarten van Smeden, Anne-Laure Boulesteix, Jennifer Catherine Camaradou, Leo Anthony Celi, Spiros Denaxas, Alastair K. Denniston, Ben Glocker, Robert M. Golub, Hugh Harvey, Georg Heinze, Michael M. Hoffman, André Pascal Kengne, Emily Lam, Naomi Lee, Elizabeth W. Loder, Lena Maier-Hein, Bilal A. Mateen, Melissa D. McCradden, Lauren Oakden-Rayner, Johan Ordish, Richard Parnell, Sherri Rose, Karandeep Singh, Laure Wynants, and Patricia Logullo. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, 2024. doi: 10.1136/bmj-2023-078378.

Sulekha Das, Avijit Kumar Chaudhuri, Nobhonil Roy Choudhury, and Partha Ghosh. Identification of the recurrence of differentiated thyroid cancer by stacking classifier. Research Square preprint, version 1, 2025.

Raymon H. Grogan, Sharone P. Kaplan, Hongyuan Cao, Roy E. Weiss, Leslie J. DeGroot, Cassie A. Simon, Omran M. A. Embia, Peter Angelos, Edwin L. Kaplan, and Rebecca B. Schechter. A study of recurrence and death from papillary thyroid cancer with 27 years of median follow-up. *Surgery*, 154(6):1436–1447, 2013. doi: 10.1016/j.surg.2013.07.008. Article pp. 1436–1446; discussion pp. 1446–1447.

Bryan R. Haugen, Erik K. Alexander, Keith C. Bible, Gerard M. Doherty, Susan J. Mandel, Yuri E. Nikiforov, Furio Pacini, Gregory W. Randolph, Anna M. Sawka, Martin Schlumberger, Kathryn G. Schuff, Steven I. Sherman, Julie Ann Sosa, David L. Steward, R. Michael Tuttle, and Leonard Wartofsky. 2015 American Thyroid Association management guidelines for adult patients with thyroid nodules and differentiated thyroid cancer: The American Thyroid Association guidelines task force on thyroid nodules and differentiated thyroid cancer. *Thyroid*, 26(1):1–133, 2016. doi: 10.1089/thy.2015.0020.

Guobin Hu, Bin Huang, Li Cai, Yue Zhang, Ying Zhang, Yuyang Liu, and Ganxun Wu. Machine learning prediction of thyroid cancer recurrence for early screening and clinical decision pathways: a retrospective cohort study. *Discover Oncology*, 17:239, 2026. doi: 10.1007/s12672-025-04293-2.

Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804.

Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4):15, 2012. doi: 10.1145/2382577.2382579. Article 15.

Mallika Khwanmuang, Watcharaporn Cholamjiak, and Pasa Sukson. Explaining risk stratification in differentiated thyroid cancer using SHAP and machine learning approaches. *Biomedicines*, 13(12):2964, 2025. doi: 10.3390/biomedicines13122964. Published 2 December 2025.

H. M. N. S. Kumari, H. M. L. S. Kumari, and U. M. M. P. K. Nawarathne. Differentiated thyroid cancer recurrence classification using machine learning models and Bayesian neural networks with varying priors: A SHAP-based interpretation of the best performing model. *International Journal of Research in Computing*, 5(2):17–42, 2026. URL `https://www.ijrcom.org/index.php/ijrc/article/view/161`.

Yi Li, Zimei Tang, Anwen Ren, Gang Tian, Jianing Zhang, Yiran Wang, Jie Liu, and Jie Ming. A machine learning-based model for predicting recurrence in intermediate- and high-risk differentiated thyroid cancer: insights from a retrospective single-center study of 2388 patients. *Frontiers in Endocrinology*, 16:1552479, 2025. doi: 10.3389/fendo.2025.1552479.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, pp. 4765–4774, 2017.

Claude Nadeau and Yoshua Bengio. Inference for the generalization error. *Machine Learning*, 52(3):239–281, 2003. doi: 10.1023/A:1024068626366.

Ahmet Cankat Öztürk, Erkan Akkur, and Serkan Çizmecioğulları. Predicting recurrence of differentiated thyroid cancer with an explainable artificial intelligence model. *Archives of Current Medical Research*, 6 (3):280–287, 2025. doi: 10.47482/acmr.1677545.

Young Min Park and Byung-Joo Lee. Machine learning-based prediction model using clinico-pathologic factors for papillary thyroid carcinoma recurrence. *Scientific Reports*, 11(1):4948, 2021. doi: 10.1038/s41598-021-84504-2.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12(85):2825–2830, 2011. URL `http://jmlr.org/papers/v12/pedregosa11a.html`.

Richard D. Riley, Joie Ensor, Kym I. E. Snell, Frank E. Harrell, Glen P. Martin, Johannes B. Reitsma, Karel G. M. Moons, Gary S. Collins, and Maarten van Smeden. Calculating the sample size required for developing a clinical prediction model. *BMJ*, 368:m441, 2020. doi: 10.1136/bmj.m441.

Keshika Shrestha, H. M. Jabed Omur Rifat, Uzzal Biswas, Jun-Jiat Tiang, and Abdullah-Al Nahid. Predicting the recurrence of differentiated thyroid cancer using whale optimization-based XGBoost algorithm. *Diagnostics*, 15(13):1684, 2025. doi: 10.3390/diagnostics15131684. Published 2 July 2025.

Deepak Thakur, Tanya Gera, Vivek Bhardwaj, R. Mazen, Ayodele Lasisi, and Trmesgen Engida. A comparative study on advanced predictive modeling of thyroid cancer recurrence using multi algorithmic machine learning frameworks. *Scientific Reports*, 16(1):3385, 2026. doi: 10.1038/s41598-025-33396-7. Published online 30 December 2025.

R. M. Tuttle, H. Tala, J. Shah, R. Leboeuf, R. Ghossein, M. Gonen, M. Brokhin, G. Omry, J. A. Fagin, and A. Shaha. Estimating risk of recurrence in differentiated thyroid cancer after total thyroidectomy and radioactive iodine remnant ablation: Using response to therapy variables to modify the initial risk estimates predicted by the new American Thyroid Association staging system. *Thyroid*, 20(12):1341–1349, 2010. doi: 10.1089/thy.2010.0178.

Ben Van Calster, David J. McLernon, Maarten van Smeden, Laure Wynants, and Ewout W. Steyerberg. Calibration: the Achilles heel of predictive analytics. *BMC Medicine*, 17(1):230, 2019. doi: 10.1186/s12916-019-1466-7. On behalf of Topic Group 'Evaluating diagnostic tests and prediction models' of the STRATOS initiative.

Daniël J. van de Berg, Pedro M. Rodriguez Schaap, Faridi S. Jamaludin, Hanneke M. van Santen, Sarah C. Clement, Menno R. Vriens, A. S. Paul van Trotsenburg, Christiaan F. Mooij, Eveline Bruinstroop, Schelto Kruijff, Robin P. Peeters, Frederik A. Verburg, Romana T. Netea-Maier, Els J. M. Nieveen van Dijkum, Joep P. M. Derikx, and Anton F. Engelsman. The definition of recurrence of differentiated thyroid cancer: A systematic review of the literature. *Thyroid*, 34(11):1324–1334, 2024. doi: 10.1089/thy.2024.0271.

Sudhir Varma and Richard Simon. Bias in error estimation when using cross-validation for model selection. *BMC Bioinformatics*, 7:91, 2006. doi: 10.1186/1471-2105-7-91.

Andrew J. Vickers and Elena B. Elkin. Decision curve analysis: A novel method for evaluating prediction models. *Medical Decision Making*, 26(6):565–574, 2006. doi: 10.1177/0272989X06295361.

Hongxi Wang, Chao Zhang, Qianrui Li, Tian Tian, Rui Huang, Jiajun Qiu, and Rong Tian. Development and validation of prediction models for papillary thyroid cancer structural recurrence using machine learning approaches. *BMC Cancer*, 24:427, 2024. doi: 10.1186/s12885-024-12146-4.

Joseph J. Wiltshire, Thomas M. Drake, Lesley Uttley, and Sabapathy P. Balasubramanian. Systematic review of trends in the incidence rates of thyroid cancer. *Thyroid*, 26(11):1541–1552, 2016. doi: 10.1089/thy.2016.0100.